Workshop
Advanced Data Modelling

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1 The Workshop

I attended a workshop on advanced data modeling; it was about making a so-called advanced Entity Relationship Diagram (ERD) and Database Design. Primary Keys, Foreign Keys, and intermediate tables were first mentioned. After that, we took a look at our current ERD, followed by how to make it more advanced, such as what factors to consider and which or what to add.

Making an ERD involves the following subsequent steps:

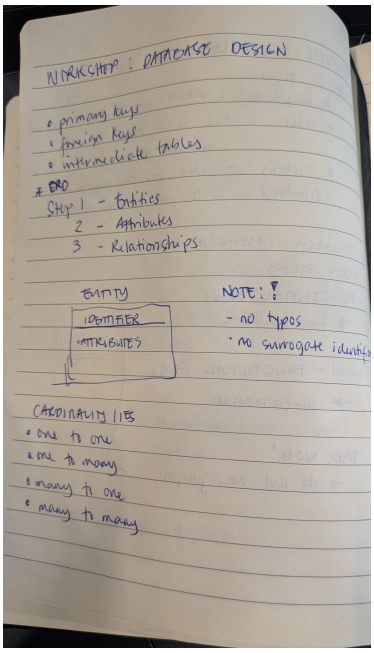
1. Identifying entities;
2. naming their attributes; and,
3. identifying the relationships between the entities.

Entities should be placed outside the box, with the identifier at the top and the common attributes listed below it. It is important to carefully check all entries to avoid typos and to ensure that no surrogate identifiers are used.

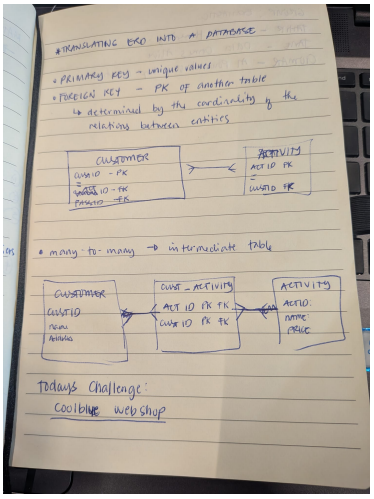
After creating the ERD, it can be transformed into a database. A database should contain:

1. Primary Keys, which are unique values that identify a single row table row; and,
2. foreign Keys, which refer to a PK of another table.

In a many-to-many relationships between tables, an intermediate table is required. Figure 1 contains the notes I have taken during the workshop.



(a) Page 1



(b) Page 2

Figure 1: Workshop Notes

2 Reflection

I gained a thorough understanding of how to create a correct ERD and identify relationships between entities, which will be very useful for my chosen topic, Process Data in Organizations. I can now create a detailed ERD in my design document to clearly represent the data model and database structure for my project.